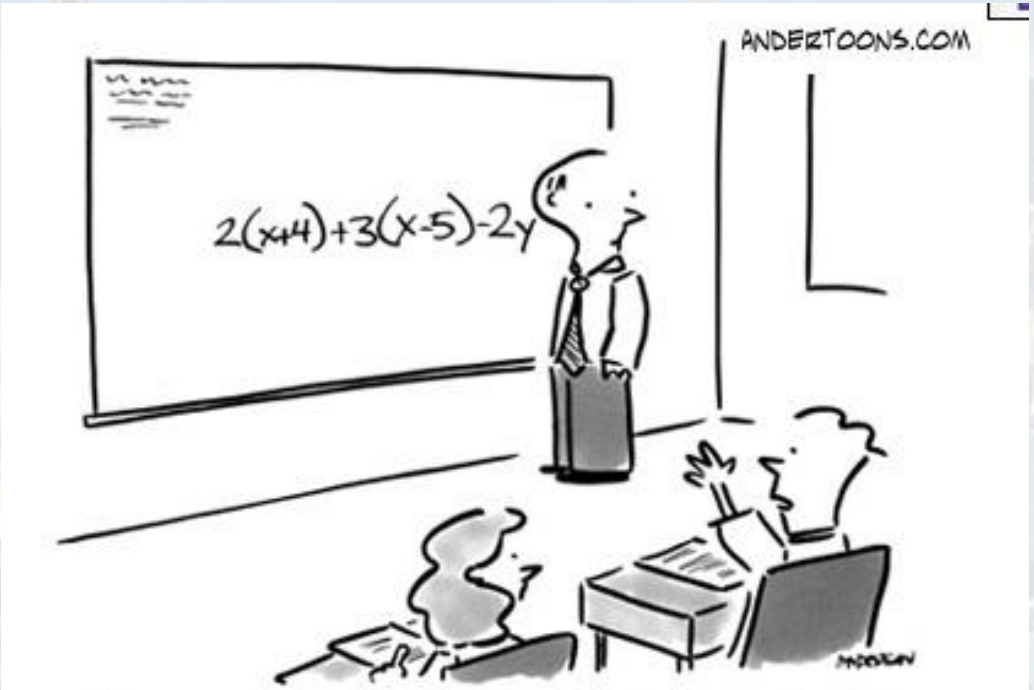
Berkeley
UNIVERSITY OF CALIFORNIA

M. Hohle:

Physics 77: Introduction to Computational Techniques in Physics



"How come we never complicate equations?"

Introduction to Computational Techniques in Physics: syllabus

Week	Date	Topic
1	June 8th	Programming Environment & UIs for Python, Programming Fundamentals
2	June 15th	Basic Types in Python
3	June 22nd	Parsing, Data Processing and File I/O, Visualization
4	June 29th	Functions, Map & Lambda
5	July 6th	Random Numbers & Probability Distributions, Interpreting Measurements
6	July 13th	Numerical Integration and Differentiation
7	July 20th	Root finding, Interpolation
8	**July 27th**	Systems of Linear Equations, **Ordinary Differential Equations (ODEs)**
9	Aug 3rd	Stability of ODEs, Examples
10	Aug 10th	Final Project Presentations

finding the intersection of **two lines**:

$$y_1 = a_1 x_1 + c_1$$

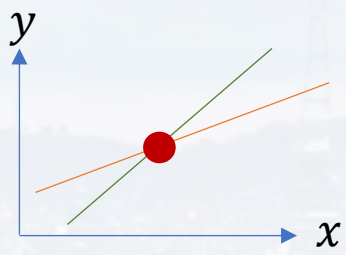
$$y_2 = a_2 x_2 + c_2$$

$$x_1 = x_2$$
$$y_1 = y_2$$

$$a_2 x + c_2 = a_1 x + c_1$$

$$x = \frac{c_2 - c_1}{a_1 - a_2}$$

$$y = a_1 \frac{c_2 - c_1}{a_1 - a_2} + c_1$$



finding the intersection of **three planes**:

$$y_1 = a_{11} x_{11} + a_{12} x_{12} + c_1$$

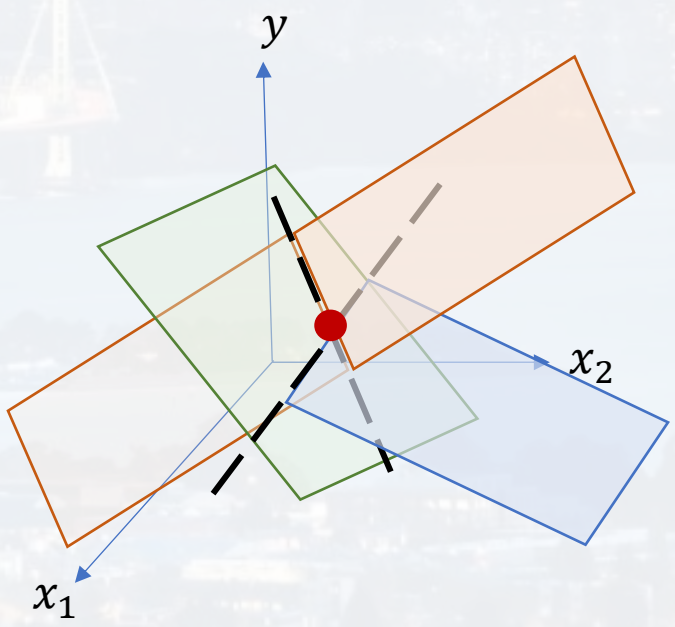
$$y_2 = a_{21} x_{21} + a_{22} x_{22} + c_2$$

$$y_2 = a_{31} x_{31} + a_{32} x_{32} + c_2$$

$$x_{11} = x_{21} = x_{31} = x_1$$

$$x_{12} = x_{22} = x_{32} = x_2$$

$$y_1 = y_2 = y_3 = y$$

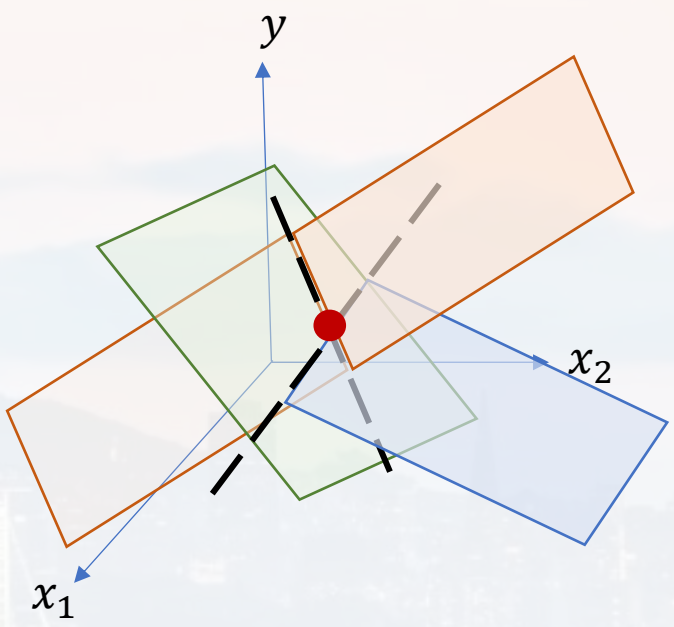


more general:

$$x_{11} = x_{21} = x_{31} = x_1$$

$$x_{12} = x_{22} = x_{32} = x_2$$

$$y_1 = y_2 = y_3 = y$$



$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n = c_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n = c_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \ldots + a_{3n}x_n = c_3$$

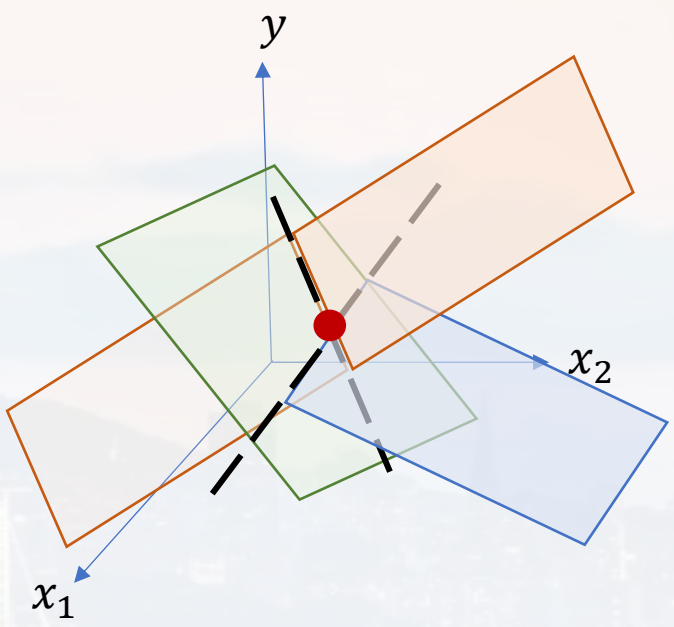
$$\ldots$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \ldots + a_{mn}x_n = c_m$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_m \end{bmatrix}$$

more general:

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_m \end{bmatrix}}_{\vec{c}}$$



$$\boxed{A\vec{x} = \vec{c}}$$

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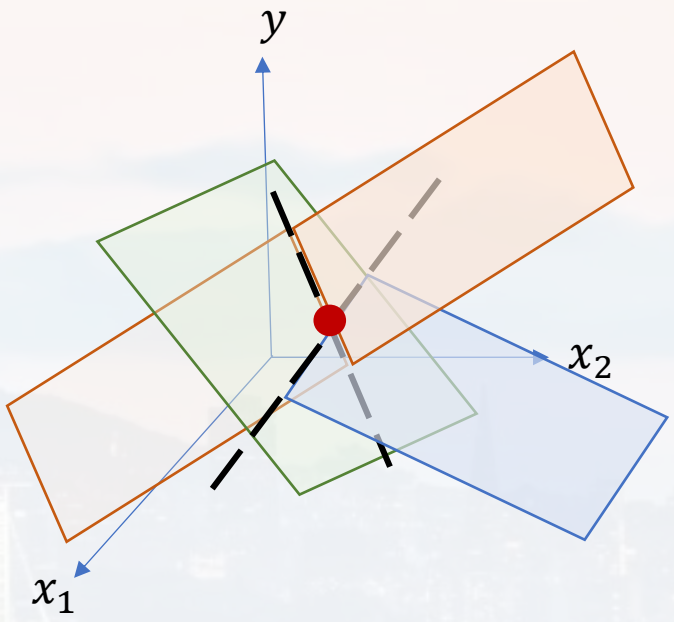

general set of solutions

for n = m → solution is unique: a point

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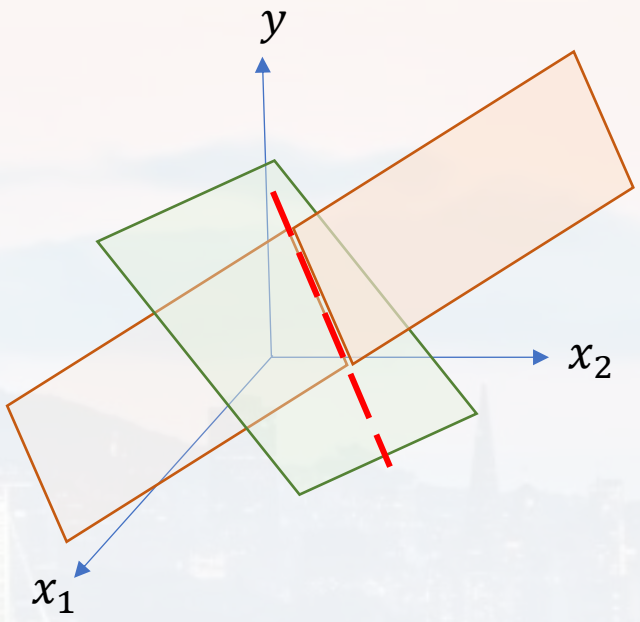

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for n = m → solution is unique: a point

for n > m (more variables than equations)
→ solution is not unique: line, hyperplane

for n < m (more equations than variables)
→ no solution

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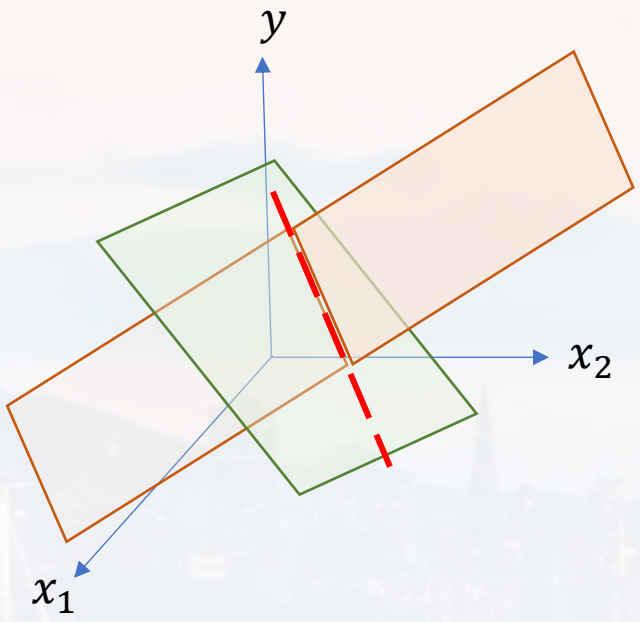

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exceptions!

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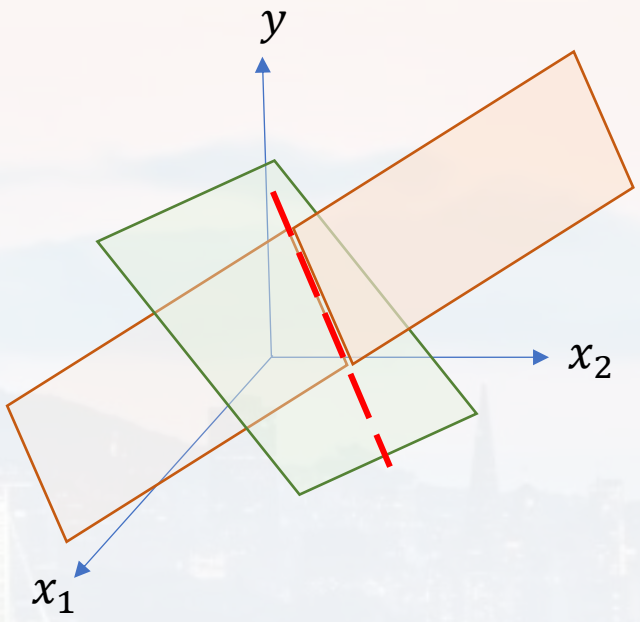

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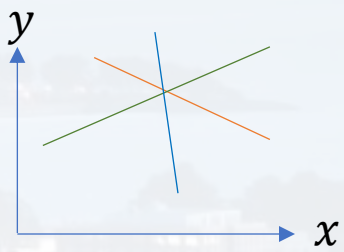

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$$\boxed{A\vec{x} = \vec{c}}$$

$\vec{x}$ = ?

for n = m

$$A^{-1}A\vec{x} = A^{-1}\vec{c}$$

$$\vec{x} = A^{-1}\vec{c}$$

$$A = [a_{ij}]$$

inverse:	$A^{-1}A = \mathrm{I}$
identity:	$I\ M = M$
transpose:	$[a_{ij}]^T = [a_{ji}]$
symmetry:	$[a_{ij}] = [a_{ji}]$
conjugate transpose:	A^+
unitary:	$A^{-1} = A^+$
idempotency:	$AA = A \rightarrow A^n = A$
normal:	$A^+A = AA^+$

solving for x:

$$\boxed{A\vec{x} = \vec{c}}$$

→ need to calculate A^{-1}
→ need to calculate a quantity called

determinant of A, *det(A)*

$$A^{-1} \sim \frac{1}{\det(A)}$$

- if $\det(A) = 0$ → no solution
- $|\det(A)|$: volume spanned by the vectors in A

inverse:	$A^{-1}A = \mathrm{I}$
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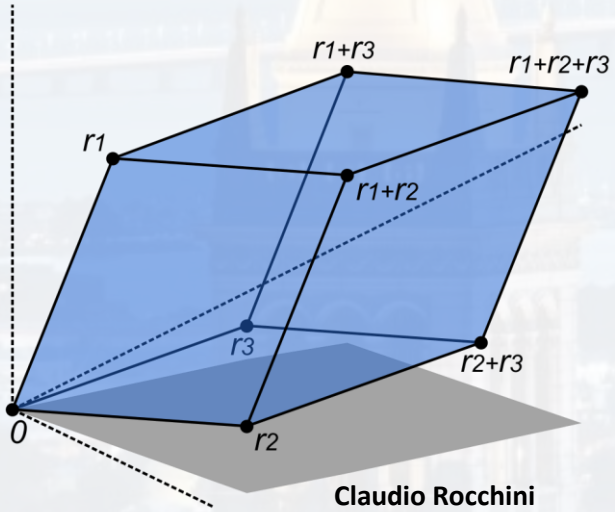
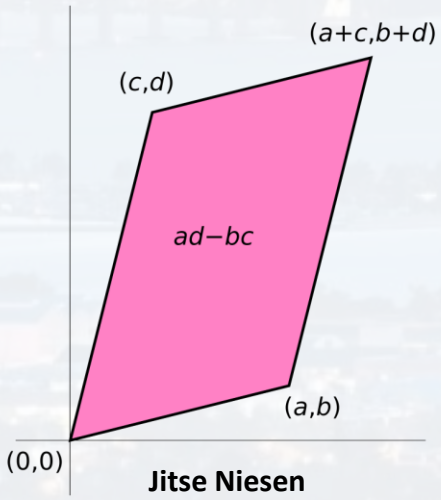


Claudio Rocchini



Jitse Niesen

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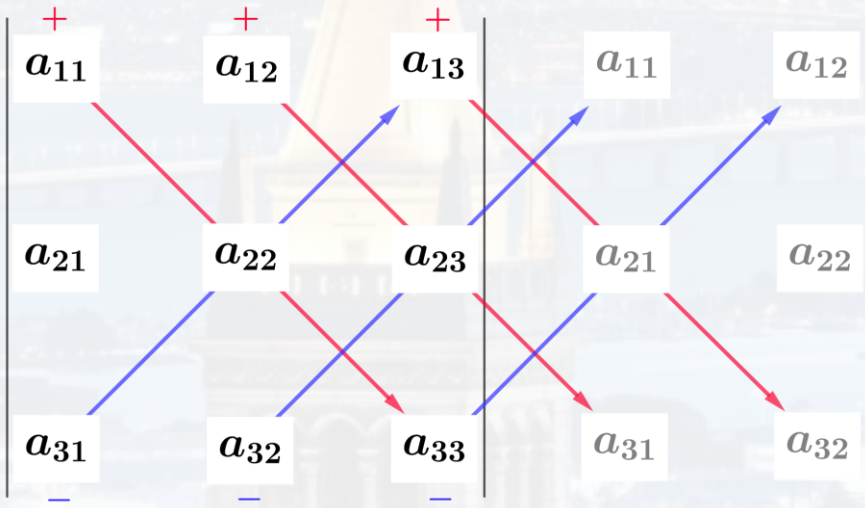

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

Kmhkmh

solving for x:

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N x N matrix:

$$\det(A) = \sum_{i_1, i_2, \ldots, i_n} \varepsilon_{i_1 \ldots i_n}\, a_{1,i_1} \ldots a_{n\, i_n} \qquad \text{where} \qquad \varepsilon_{i_1 \ldots i_n} = \boxed{\prod_{1 \le \mu < \vartheta \le n} \operatorname{sgn}(i_\vartheta - i_\mu)}$$

(Levi-Civita symbol)

changing indices does not change $|\det(A)|$

determinant of A, *det(A)*

$$\boxed{A\vec{x} = \vec{c}}$$

inverse:	$A^{-1}A = \mathrm{I}$
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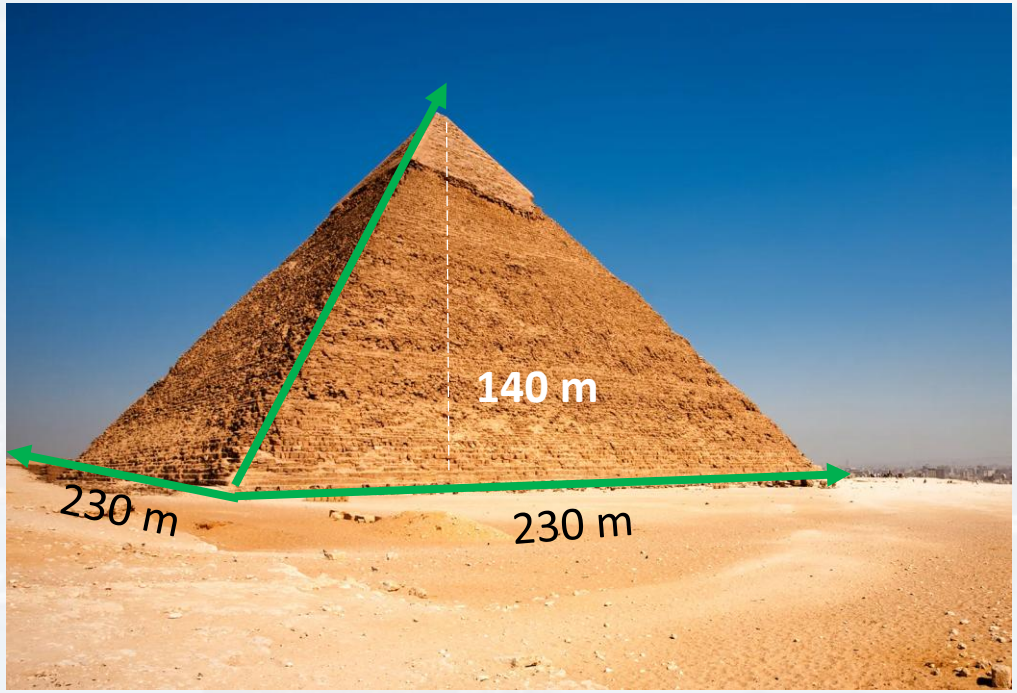


$$\varepsilon_{i_1 \ldots i_n} = \prod_{1 \le \mu < \vartheta \le n} \mathrm{sgn}(i_\vartheta - i_\mu)$$

$$V = \left| \det \begin{pmatrix} 230 & 0 & 115 \\ 0 & 230 & 115 \\ 0 & 0 & 140 \end{pmatrix} \right| \frac{1}{3} = \frac{230 * 230 * 140 + 0 + 0\ -\ \ 0\ - 0 - 0}{3} = 2{,}468{,}666\ m^3$$

determinant of A, *det(A)*

$$\varepsilon_{i_1 \ldots i_n} = \prod_{1 \le \mu < \vartheta \le n} \operatorname{sgn}(i_\vartheta - i_\mu)$$

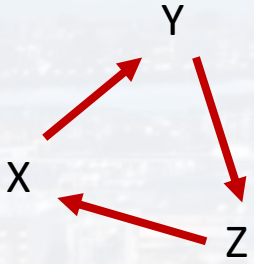
$$V = \left| \det \begin{pmatrix} 230 & 0 & 115 \\ 0 & 230 & 115 \\ 0 & 0 & 140 \end{pmatrix} \right| \frac{1}{3} = \frac{230 * 230 * 140 + 0 + 0 \; - \; 0 \; - 0 - 0}{3} = 2{,}468{,}666 \; m^3$$

volume does not depend on **where** I put my **coord origin**…

$$V = \left| \det \begin{pmatrix} 0 & 230 & 115 \\ 230 & 0 & 115 \\ 0 & 0 & 140 \end{pmatrix} \right| \frac{1}{3} = \left| \frac{0 + 0 + 0 \; - \; 140 * 230 * 230 \; - 0 - 0}{3} \right| = 2{,}468{,}666 \; m^3$$

…or how I **turn** the object!

$$V = \left| \det \begin{pmatrix} 115 & 230 & 0 \\ 115 & 0 & 230 \\ 140 & 0 & 0 \end{pmatrix} \right| \frac{1}{3} = \left| \frac{0 + 230 * 230 * 140 + 0 \; - \; 0 - 0 - 0}{3} \right| = 2{,}468{,}666 \; m^3$$



$$V = \left| \det \begin{pmatrix} 140 & 0 & 0 \\ 115 & 230 & 0 \\ 115 & 0 & 230 \end{pmatrix} \right| \frac{1}{3} = \left| \frac{140 * 230 * 230 + 0 + 0 \; - \; 0 - 0 - 0}{3} \right| = 2{,}468{,}666 \; m^3$$

linear regression

Goal 1: finding a model that tells us how we can **predict y_k from all the x_i**

	x_1	x_2	x_3	x_4	x_5	y_k
Index	molecular_weight	electronegativity	bond_lengths	num_hydrogen_bonds	logP	toxicity_score
0	341.704	2.65585	3.09407	2	9.11147	80.9281
1	335.951	3.22262	2.89039	7	8.92848	83.4911
2	235.203	2.44115	2.48203	1	6.49731	61.8406
3	246.505	2.76656	2.71547	7	7.45089	57.0538
4	437.939	3.4801	3.59569	3	10.9156	131.326
	336.453	2.81474	3.11	9	8.55696	?

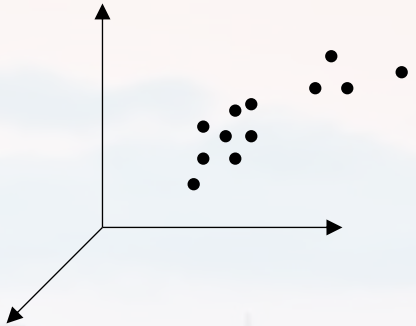
kth row

Goal 2: once we have a model: **predict y_k from new data set**

linear regression

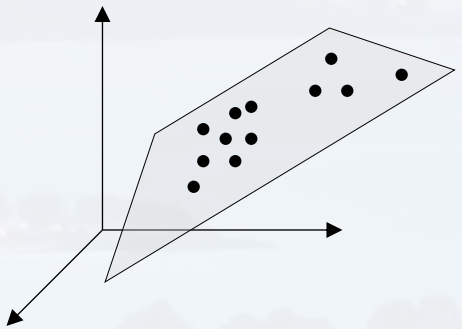
<u>idea:</u> data point y_k in N dimensional space

→ $y_k = f(x_1, \dots x_n, \dots x_N) + \epsilon$ for each data point k



<u>ansatz:</u> $\boxed{y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon}$ ***linear*** combination

y: response
x: regressors **(assumed to be independent)**
β: factors **(how a regressor contributes to the response)**
β_0: intercept
ε: error **(stochasticity of the data, assumed to be normally dist.)**



Finding β_n is the model!

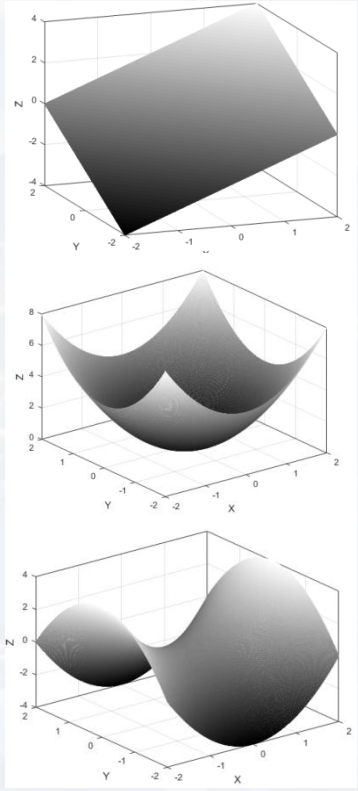
y:	**response**
x:	**regressors**
β:	**factors**
$\boldsymbol{\beta_0}$**:**	**intercept**
ε:	**error**

general: linear refers to the **factors**

$$y_k = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$ 2*D* plane in 3*D* space

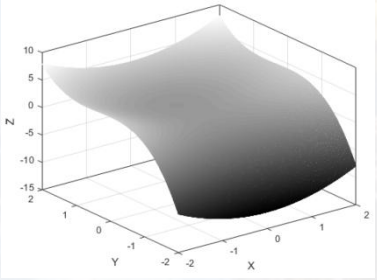
$$y_k = \beta_0 + \beta_1 {x_1}^2 + \beta_2 {x_2}^2$$ 2*D* parabolic

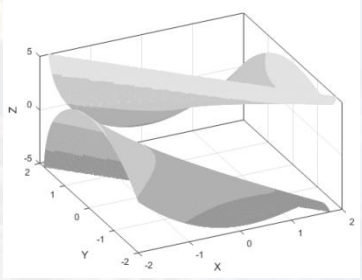
$$y_k = \beta_0 + \beta_1 {x_1}^2 - \beta_2 {x_2}^2$$ 2*D* hyperbolic



all linear

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

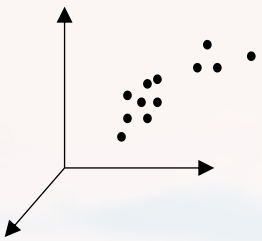




...and many more...

for *K* data points in *N* dimensional space

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$



y:	**response**
x:	**regressors**
β:	**factors**
β_0:	**intercept**
ε:	**error**

$$\underbrace{\begin{pmatrix} y_1 \\ \dots \\ y_k \\ \dots \\ \dots \\ y_K \end{pmatrix}}_{Y} = \underbrace{\begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1n} & \dots & x_{1N} \\ \dots & \dots & & & \dots & & \\ 1 & x_{k1} & & & x_{kn} & & \\ 1 & \dots & & & \dots & & \\ 1 & x_{K1} & x_{K2} & \dots & x_{Kn} & \dots & x_{KN} \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \dots \\ \beta_n \\ \dots \\ \beta_N \end{pmatrix}}_{\beta} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_k \\ \dots \\ \varepsilon_K \end{pmatrix}}_{\varepsilon}$$

$$\boxed{Y = X\beta + \varepsilon}$$

<u>fitting:</u> finding the best β in terms of minimizing the errors

$$(Y - X\beta)^T (Y - X\beta) = \sum_k {\varepsilon_k}^2$$

$$\frac{\partial}{\partial \beta} \sum_k {\varepsilon_k}^2 = 0 \longrightarrow \beta_{best} = \hat{\beta} = (X^T X)^{-1} X^T Y \longrightarrow \widehat{\boldsymbol{Y}} = \boldsymbol{X}\widehat{\boldsymbol{\beta}} = X(X^T X)^{-1} X^T Y$$

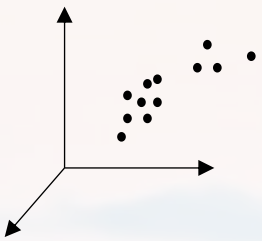
the model

X and Y are all observables

for K data points in N dimensional space

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

$$\boxed{Y = X\beta + \varepsilon}$$



y:	**response**
x:	**regressors**
β:	**factors**
β_0:	**intercept**
ε:	**error**

check out `Walk_Through_LinRegression.ipynb`

predicting **toxicity of molecules** based on their features

Index	molecular_weight	electronegativity	bond_lengths	num_hydrogen_bonds	logP	toxicity_score
0	341.704	2.65585	3.09407	2	9.11147	80.9281
1	335.951	3.22262	2.89039	7	8.92848	83.4911
2	235.203	2.44115	2.48203	1	6.49731	61.8406
3	246.505	2.76656	2.71547	7	7.45089	57.0538
4	437.939	3.4801	3.59569	3	10.9156	131.326

see `Walk_Through_LinRegression.ipynb`

```
import numpy as np

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import pylab
import scipy.stats as stats
import statsmodels.api as sm

from statsmodels.formula.api import ols
from sklearn.preprocessing import MinMaxScaler
```

- `pandas as pd` → reading .xlsx .csv .txt ...
- `matplotlib.pyplot as plt` → standard plots
- `seaborn as sns` → fancy plots: here a pair-plot
- `pylab` → Q-Q plot
- `scipy.stats as stats` → Q-Q plot
- `statsmodels.api as sm` → the actual super tool for superb data analysis
- `statsmodels.formula.api import ols` → the actual super tool for superb data analysis
- `sklearn.preprocessing import MinMaxScaler` → scaling and normalizing

1) loading data
2) plotting data
3) scaling data
4) fitting model
5) evaluating model

```
Train  = pd.read_csv("molecular_train_gbc.csv")

Test   = pd.read_csv("molecular_test_gbc.csv")
```

	x_1	x_2	x_3	x_4	x_5	y_k
Index	molecular_weight	electronegativity	bond_lengths	num_hydrogen_bonds	logP	toxicity_score
0	341.704	2.65585	3.09407	2	9.11147	80.9281
1	335.951	3.22262	2.89039	7	8.92848	83.4911
2	235.203	2.44115	2.48203	1	6.49731	61.8406
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4	437.939	3.4801	3.59569	3	10.9156	131.326

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

y: toxicity_score

x_n : molecular_weight, electronegativity, bond_lengths, num_hydrogen_bonds, logP

1) loading data
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3) scaling data
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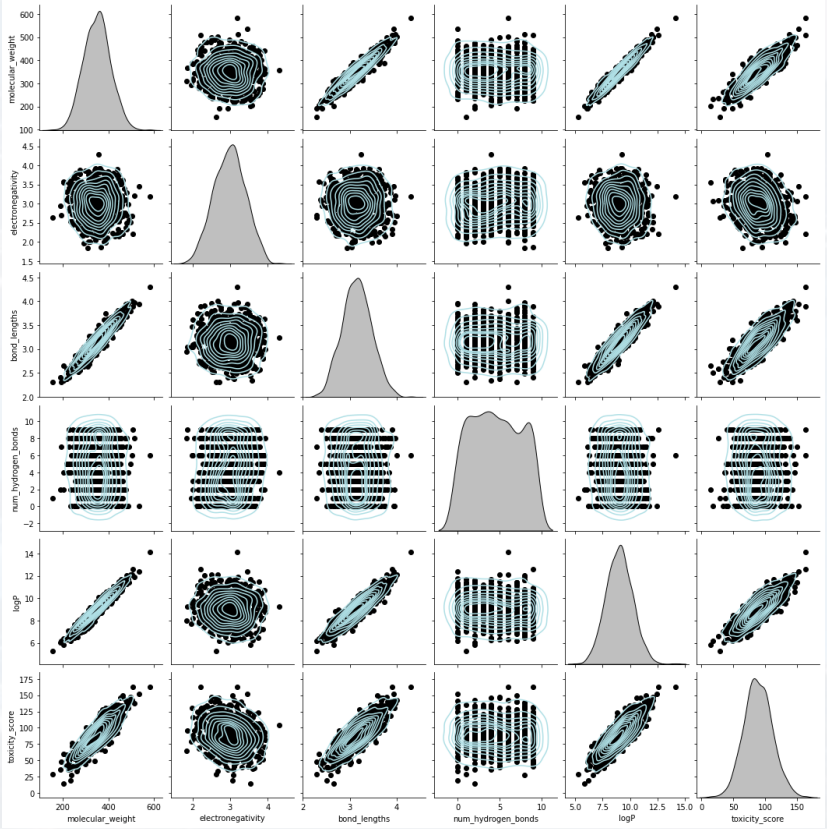
5) evaluating model

```
Train  = pd.read_csv("molecular_train_gbc.csv")
Test   = pd.read_csv("molecular_test_gbc.csv")

out = sns.pairplot(Train, kind = "kde", \
                   plot_kws = {'color':[176/255, 224/255, 230/255]},\
                   diag_kws = {'color': 'black'})
out.map_offdiag(plt.scatter, color = 'black')
```



1) loading data
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                   diag_kws = {'color': 'black'})
out.map_offdiag(plt.scatter, color = 'black')
```

Scaling the data because unit system is arbitrary!

Large numerical values dominate the optimization! → rescaling!

molecular_weight	electronegativity	bond_lengths
341.704	2.65585	3.09407

```
scaler  = MinMaxScaler(feature_range = (0, 1))
TrainS  = scaler.fit_transform(Train)
TestS   = scaler.transform(Test)
```

the scaler returns an `np.array`
→ convert back to `data frame`

```
TrainS = pd.DataFrame(TrainS, columns = Train.columns)
TestS  = pd.DataFrame(TestS, columns = Train.columns)
```

1) loading data
2) plotting data
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```
TrainS = pd.DataFrame(TrainS, columns = Train.columns)
TestS  = pd.DataFrame(TestS,  columns = Train.columns)
```

```
equation = 'toxicity_score ~ ' + '+'.join(Train.columns[:-1])
print(equation)
```

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

```
toxicity_score ~      molecular_weight + electronegativity +
                      bond_lengths + num_hydrogen_bonds + logP
```

```
my_model = ols(equation, data = TrainS).fit()
my_model.summary()
```

OLS (ordinary least squares)

1) loading data
2) plotting data
3) scaling data
4) fitting model
5) evaluating model

```
my_model.summary()
```

not the fit quality!

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          toxicity_score   R-squared:                       0.790
Model:                             OLS   Adj. R-squared:                  0.789
Method:                  Least Squares   F-statistic:                     597.5
Date:                 Fri, 13 Sep 2024   Prob (F-statistic):          3.34e-266
Time:                         20:57:10   Log-Likelihood:                 1013.0
No. Observations:                  800   AIC:                            -2014.
Df Residuals:                      794   BIC:                            -1986.
Df Model:                            5
Covariance Type:             nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
Intercept             0.1494      0.012     12.533      0.000       0.126       0.173
molecular_weight      0.7961      0.089      8.982      0.000       0.622       0.970
electronegativity    -0.1682      0.015    -11.591      0.000      -0.197      -0.140
bond_lengths          0.0204      0.049      0.417      0.677      -0.076       0.116
num_hydrogen_bonds    0.0035      0.008      0.458      0.647      -0.011       0.018
logP                  0.1246      0.072      1.723      0.085      -0.017       0.267
==============================================================================
Omnibus:                        2.249   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.325   Jarque-Bera (JB):                2.240
Skew:                          -0.129   Prob(JB):                        0.326
Kurtosis:                       2.980   Cond. No.                         65.6
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

p-value for constant model

number of data points is much larger than the number of regressors
→ degree of freedom approx. no of obs

p-values for factors (should be < 0.01)

2σ conf range of factors

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

more accurate: determining **the p-values for the factors using ANOVA** for the corresponding residuals

```
table     = sm.stats.anova_lm(my_model, typ = 1)
print(table)
```

1) loading data
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$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

	df	sum_sq	mean_sq	F	PR(>F)
molecular_weight	1.0	13.346285	13.346285	2847.525516	8.024085e-265
electronegativity	1.0	0.640388	0.640388	136.631363	3.085962e-29
bond_lengths	1.0	0.000684	0.000684	0.145954	7.025342e-01
num_hydrogen_bonds	1.0	0.000703	0.000703	0.150055	6.985866e-01
logP	1.0	0.013917	0.013917	2.969353	8.524510e-02
Residual	794.0	3.721459	0.004687	NaN	NaN

vs from t-test

0.0000
0.0000
0.6766
0.6473
0.0852

```
residuals = my_model.resid

plt.hist(residuals, color = 'w', edgecolor = 'black')
plt.title('fit residuals')
plt.ylabel('#')
plt.xlabel('value')
plt.show()
```

1) loading data
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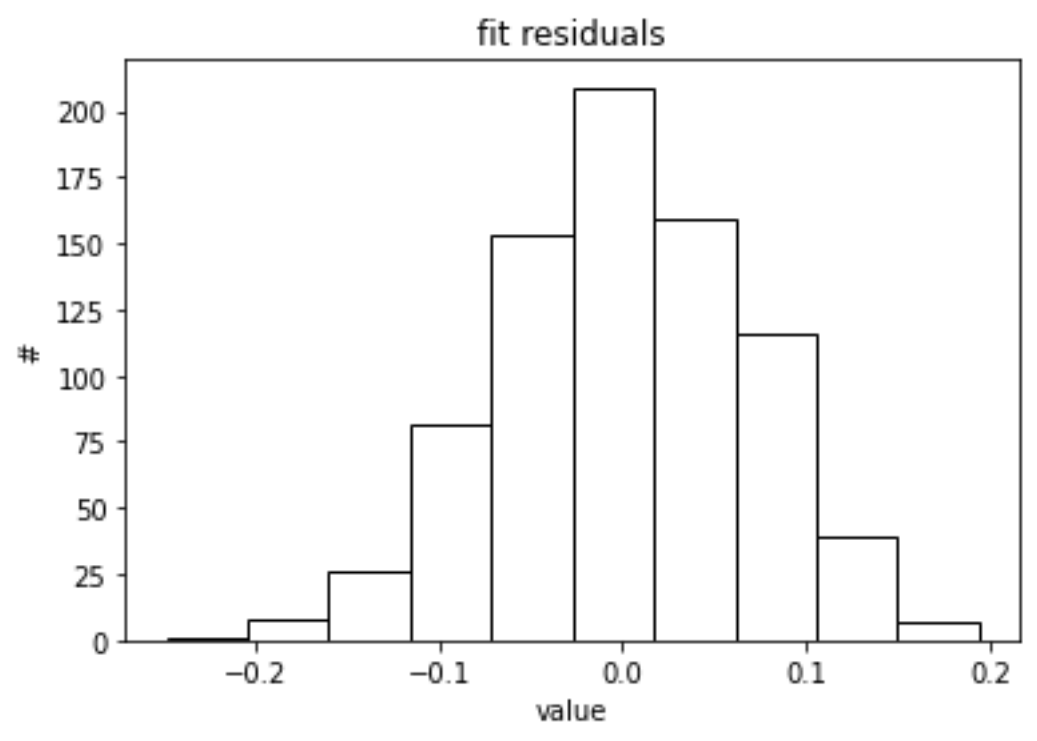
$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$



residuals approx. normally distributed around μ = 0

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5) evaluating model

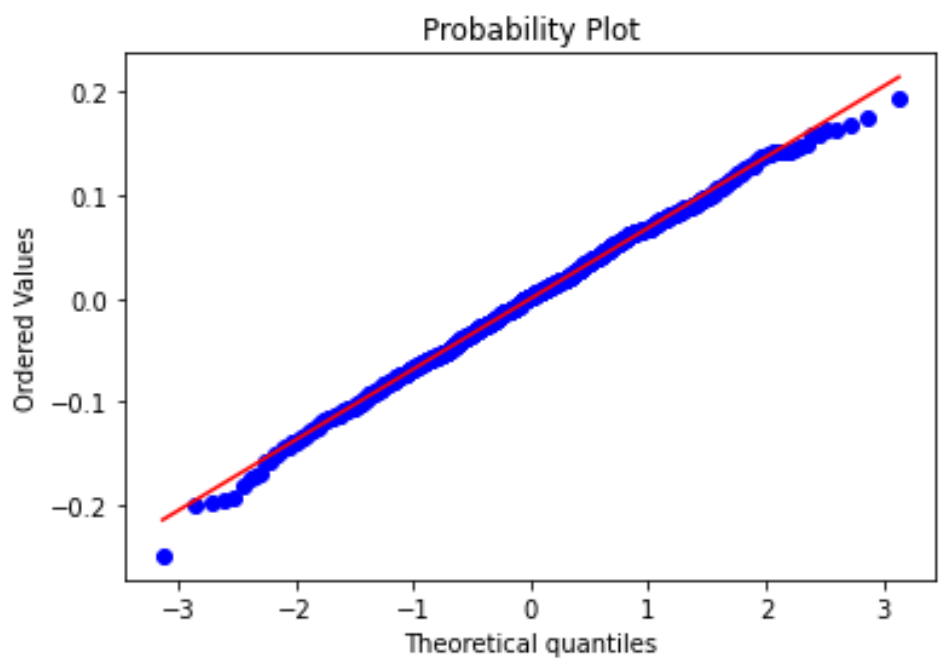
```
residuals = my_model.resid

plt.hist(residuals, color = 'w', edgecolor = 'black')
plt.title('fit residuals')
plt.ylabel('#')
plt.xlabel('value')
plt.show()

stats.probplot(residuals, dist = "norm", plot = pylab)
pylab.show()
```

residuals approx. normally distributed around μ = 0

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$



```
Ypred  = my_model.predict(TestS)

higher = np.max([Ypred, TestS.toxicity_score])
lower  = np.min([Ypred, TestS.toxicity_score])

plt.plot([lower, higher], [lower, higher], c = [0, 0, 0, 0.2],\
         linewidth = 4)
plt.scatter(TestS.toxicity_score, Ypred, marker = '.', c = 'k')
plt.ylabel('prediction')
plt.xlabel('toxicity score')
plt.show()
```

1) loading data
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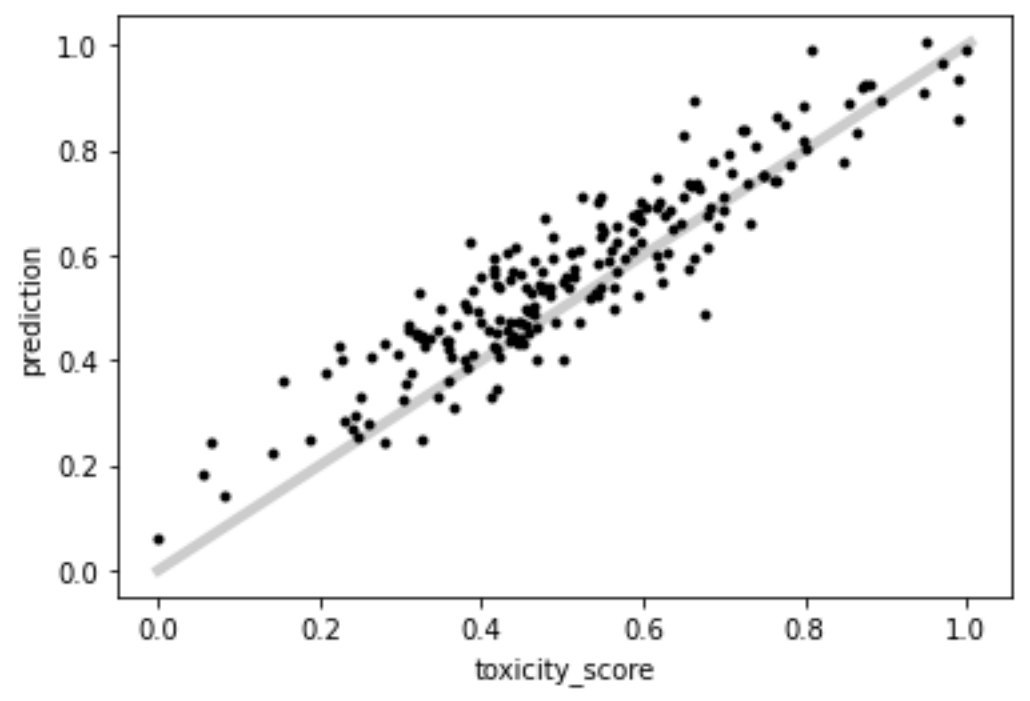
$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$



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plt.show()
```
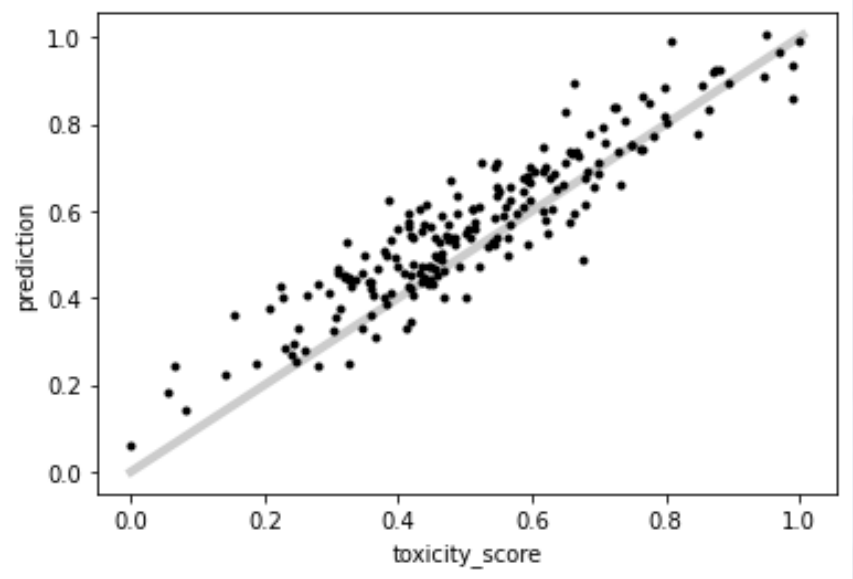
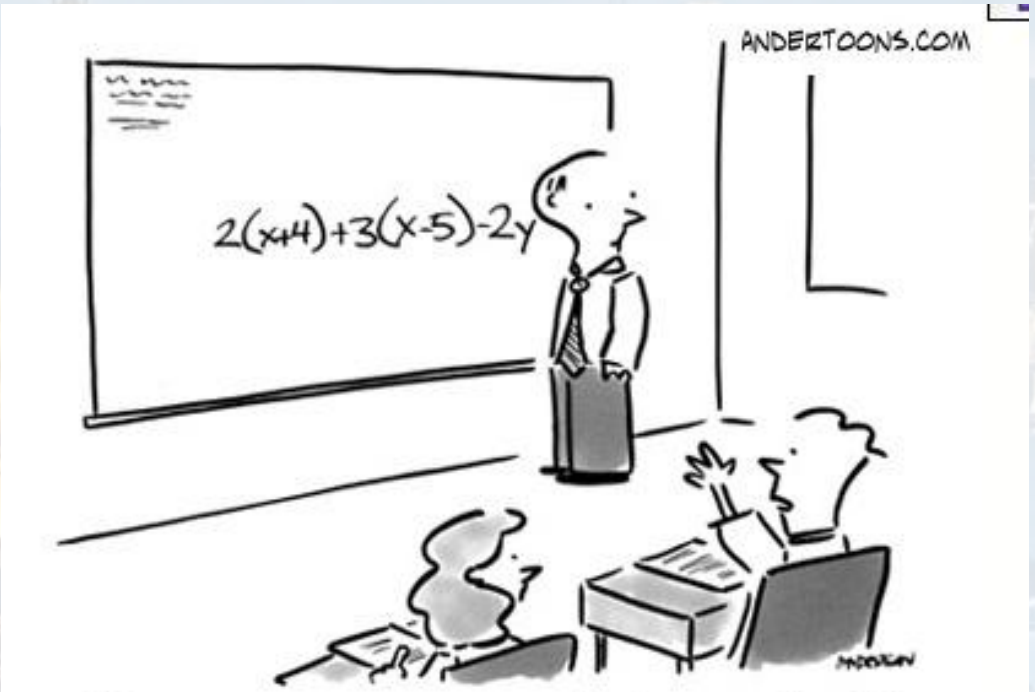




```
mean_dev = np.sum( abs(TestS.toxicity_score - Ypred) )/len(Ypred)
print(mean_dev)
```

5%

M. Hohle:

Thank you for your attention!



"How come we never complicate equations?"